STIVEN CATALYST

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MANAGEMENT REVIEW

Management without theatre.



STRATEGY

OPERATIONAL RISK: FMEA AND THE RISK MATRIX

Why the RPN takes only 120 values, two teams and one process, the Action Priority of 2019, what is wrong with risk matrices, and a card for failure modes.

120 different values are all the RPN can take, out of 1,000 combinations of severity, occurrence and detection.

BOWLES, 2003

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Every operation keeps a list of what could go wrong. How that list gets scored decides which risk is tackled first. The most common scores mislead; here is why, and what to use instead.

FMEA began in the US military in 1949. Its Risk Priority Number multiplies three scores from 1 to 10, yet its 1,000 combinations give only 120 different values, and a severe failure can rank below a mild one. When two hospital teams analysed the same process, only 17 of their 50 failures each were the same. The 2019 AIAG & VDA handbook replaced the number with an Action Priority, and Tony Cox showed that risk matrices can rank smaller risks above larger ones.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editor interprets rather than the research, it says so.

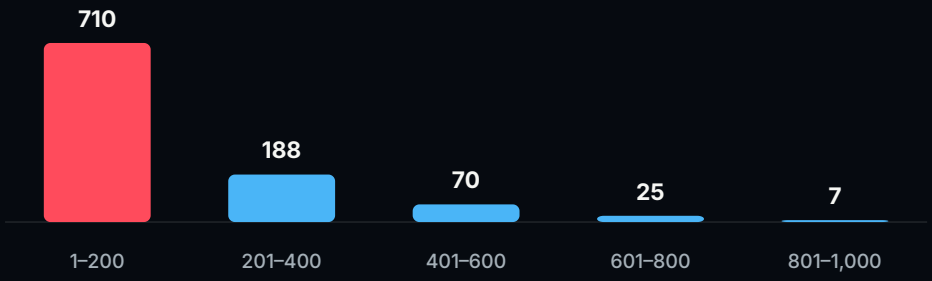
Practice
The steps and the card are proposals to try, not research results.

COVER STORY

120 out of 1,000

Failure Mode and Effects Analysis began in the US military, with procedure MIL-P-1629 of 1949. Its best-known number is the Risk Priority Number: severity × occurrence × detection, each scored from 1 to 10.

HOW THE 1,000 COMBINATIONS OF THE THREE SCORES SPREAD OVER THE RPN (EDITORS' CALCULATION)



The RPN looks like a scale from 1 to 1,000, but John Bowles showed that only 120 different values can occur. None lies between 901 and 999, and 120 itself comes from 24 different combinations. A severe failure can rank low: severity 10 with occurrence 1 and detection 1 gives 10, while 4 × 4 × 4 gives 64.

OUR READING

Multiplying three ordinal scores produces a number that looks more precise than it is. Read one by one, the three scores say more.

Sources: U.S. Armed Forces, 1949 (via UK MOD ASEMS toolkit); John B. Bowles, Annual Reliability and Maintainability Symposium (IEEE), 2003 (via Produção, 2013)

The chart and the 24 ways to reach 120 are the editors' calculation over all 1,000 combinations; the 120 values follow Bowles.

THE NUMBERS

Two teams, one process

In two hospitals of the same NHS trust, two teams ran an FMEA of the same process at the same time: the use of the antibiotics vancomycin and gentamicin.

FAILURES FOUND BY EACH TEAM, AND BY BOTH



Their scores for severity and detectability, and their RPNs, differed markedly, so the failures were ranked differently. In a study of hazards in public leisure activities, David Ball and John Watt found that different assessors placed the same hazard very differently on a risk matrix, and the spread stayed large even after long reflection.

OUR READING

An FMEA tells you as much about the team as about the process. What one team misses, a second team or a second look will often find.

Sources: Nada Atef Shebl, Bryony Dean Franklin & Nick Barber, Journal of Patient Safety, 2009 (via Abstract); David Ball & John Watt, Risk Analysis, 2013 (via Abstract)

17 of 50 is 34% for each team; the study's 17% is counted over all 100 failures. One process in one trust.

THE MODEL

From RPN to Action Priority

In June 2019 AIAG and VDA, the US and German automotive bodies, published a joint FMEA handbook. It sets out seven steps and replaces the RPN with an Action Priority, read from a table that gives severity the most weight, then occurrence, then detection.

Analysis

1 Planning and preparation

2 Structure analysis

3 Function analysis

4 Failure analysis

Risk and action

5 Risk analysis

6 Optimisation

7 Documenting the results (new)

01	**High**	Highest priority: improve prevention or detection, or document why the current controls are enough.
02	**Medium**	Find actions or, if the company decides so, document why the controls are enough.
03	**Low**	Actions may be found.

OUR READING

The question moves from “how big is the number?” to “what must we do?”, and a team can act on the second one.

Source: AIAG & VDA QMC, 2019 (via Publisher pages and Relyence guide)

The grouping of the steps is the editors'. Some customers still accept the RPN alongside the Action Priority.

More in the essay Why the shift handover is one of the most underrated processes

WHAT THE RESEARCH SAYS

What is wrong with risk matrices

In 2008 Tony Cox analysed risk matrices mathematically in the journal Risk Analysis. He found four kinds of problem.

01	**Poor resolution**	Typical matrices compare correctly only a small share of pairs of risks, in his example less than 10%, and give the same rating to very different risks.
02	**Errors**	They can rate smaller risks above larger ones; when frequency and severity are negatively related, they can be worse than useless.
03	**Resources**	Resources for reducing risk cannot be allocated well on the basis of their categories.
04	**Judgment**	Inputs and outputs need subjective interpretation; different users can rate the same risks in opposite ways.

In 2013 Philip Thomas, Reidar Bratvold and Eric Bickel found no published scientific study showing that risk matrices improve risk decisions. In one drilling example, reversing the scoring scale changed the order of priorities.

OUR READING

A matrix is a picture of judgments. When the team looks at the risk ranked highest, whose judgment is it looking at?

Sources: Louis Anthony (Tony) Cox Jr., Risk Analysis, 2008 (via Abstract); Philip Thomas, Reidar B. Bratvold & J. Eric Bickel, SPE Annual Technical Conference and Exhibition, 2013 (via JPT summary, 2014; Oil IT Journal)

Cox's results are mathematical, not measurements in organisations; the 10% is his example.

HOW IT IS MEASURED

Scoring risk without fooling yourself

ISO 31000 defines risk as the effect of uncertainty on objectives, positive or negative, and treats managing it as a process: identify, analyse, evaluate, treat, and keep monitoring and communicating.

1. **List failures with the people who do the work**
   Two groups, or two sessions, find more than one.
2. **Keep the three scores apart**
   Write down severity, occurrence and detection, not only their product.
3. **Look at severity first**
   A failure with severity 9 or 10 gets an action or a written reason, whatever its RPN.
4. **Score again after the action**
   If nothing was done, the risk has not changed.

HYPOTHETICAL EXAMPLE, TWO FAILURES IN A WAREHOUSE

Label	Wrong label on a pallet: 4 × 6 × 3 = RPN 72
Cold chain	A medicine order left unrefrigerated: 10 × 2 × 3 = RPN 60

The RPN puts the label first; severity puts the cold chain first. Both failures and their scores are made up.

Source: ISO, 2018 (via Excerpts, through web search)

The steps and the example are the editors'; the scores are severity × occurrence × detection.

TOOL OF THE ISSUE

The failure-mode card

One card per process step. Fill it in with the people who do the work, and keep the three scores side by side.

01 Process step where in the process, and who does it

02 What can go wrong the failure mode, in the words of the people who see it

03 Effect on the customer, the next step or safety

04 Severity · occurrence · detection three scores from 1 to 10, written separately

05 Priority high, medium or low, and the reason

06 Action and check what we do, who, by when, and the scores afterwards

Source: AIAG & VDA QMC, 2019 (via Publisher pages and Relyence guide)

A practice proposed by the editors, after the AIAG & VDA handbook (2019).

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

MIL-P-1629: Procedures for Performing a Failure Mode, Effects and Criticality Analysis
U.S. Armed Forces, 1949 · via UK MOD ASEMS toolkit

An Assessment of RPN Prioritization in a Failure Modes Effects and Criticality Analysis
John B. Bowles, Annual Reliability and Maintainability Symposium (IEEE), 2003 · via Produção, 2013

Is failure mode and effect analysis reliable?
Nada Atef Shebl, Bryony Dean Franklin & Nick Barber, Journal of Patient Safety, 2009 · via Abstract
https://doi.org/10.1097/PTS.0b013e3181a6f040

Further Thoughts on the Utility of Risk Matrices
David Ball & John Watt, Risk Analysis, 2013 · via Abstract
https://doi.org/10.1111/risa.12057

AIAG & VDA FMEA Handbook: Design FMEA, Process FMEA, Supplemental FMEA for Monitoring & System Response
AIAG & VDA QMC, 2019 · via Publisher pages and Relyence guide
https://webshop.vda.de/QMC/en/aiag-vda-fmea-handbook_eng

What's Wrong with Risk Matrices?
Louis Anthony (Tony) Cox Jr., Risk Analysis, 2008 · via Abstract
https://doi.org/10.1111/j.1539-6924.2008.01030.x

The Risk of Using Risk Matrices
Philip Thomas, Reidar B. Bratvold & J. Eric Bickel, SPE Annual Technical Conference and Exhibition, 2013 · via JPT summary, 2014; Oil IT Journal
https://doi.org/10.2118/166269-MS

ISO 31000:2018 Risk management — Guidelines
ISO, 2018 · via Excerpts, through web search
https://www.iso.org/standard/65694.html

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what it measures, in the full text where it could be opened. A source read only in a summary, an abstract or excerpts is marked “via”. A figure not confirmed in what was read is marked so on its page, or left out. The editor's interpretation is marked “Our reading”.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

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Edited by Stiven Janaqi, who answers for it; “the editors” in this publication means him. Its texts, translations, layout and the checking of figures against their sources are done in working sessions with Claude, an AI model by Anthropic.
stivencatalyst.com/how-this-is-made.html